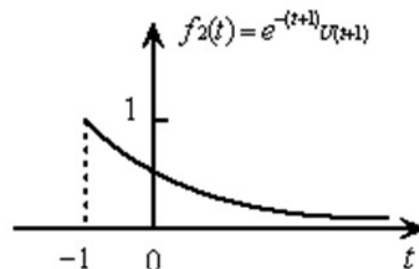
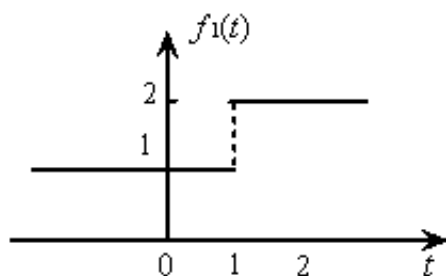




课下作业

5-1 求 $y(t) = f_1(t) * f_2(t)$ ，并画出 $y(t)$ 的波形



5-2 证明自相关函数的性质：

- 1) $R_x(0)$ 为自相关函数的最大值： $R_x(0) \geq R_x(\tau)$
- 2) 自相关函数的取值范围： $\mu_x^2 - \sigma_x^2 \leq R_x(\tau) \leq \mu_x^2 + \sigma_x^2$
- 3) $R_x(-\tau) = R_x(\tau)$ $R_x(\tau)$ 是 τ 的偶函数

5-3 证明互相关函数的性质：

- 1) 互相关函数的取值范围： $(\mu_x \mu_y - \sigma_x \sigma_y) \leq R_{xy}(\tau) \leq (\mu_x \mu_y + \sigma_x \sigma_y)$
- 2) $R_{xy}(-\tau) = R_{yx}(\tau)$



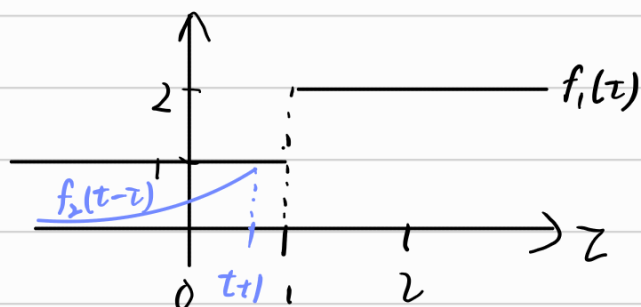
5-1

$$f_1(t) = 1 + V(t-1)$$

$$f_2(t) = e^{-(t+1)} V(t+1)$$

$$y(t) = \int_{-\infty}^{+\infty} f_1(\tau) f_2(t-\tau) d\tau = f_1(t) * f_2(t)$$

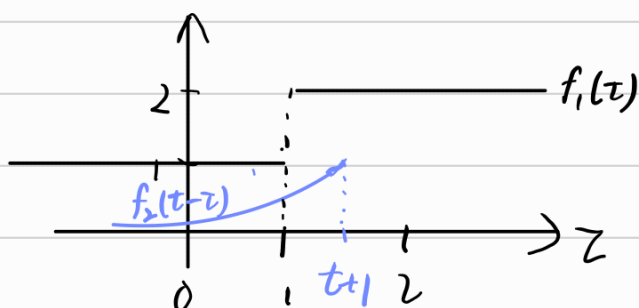
当 $t \leq 0$ 时, $f_1(\tau)$ 和 $f_2(t-\tau)$ 的图如下:



$$\begin{aligned} y(t) &= f_1(t) * f_2(t) \\ &= \int_{-\infty}^{+\infty} f_1(\tau) f_2(t-\tau) d\tau \\ &= \int_{-\infty}^{t+1} 1 \cdot e^{-(t-\tau+1)} d\tau \\ \text{let } u &= \tau - t - 1 \quad \frac{du}{d\tau} = 1 \Rightarrow du = d\tau \end{aligned}$$

$$\int_{-\infty}^0 e^u du = 1$$

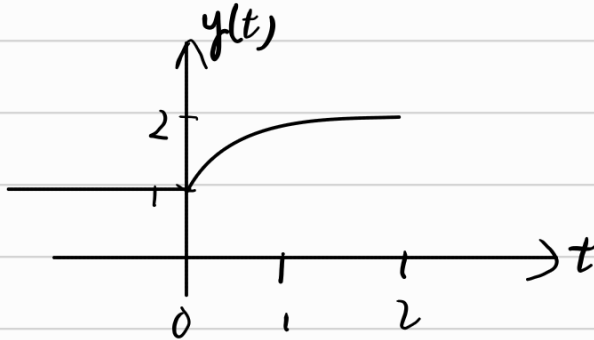
当 $t > 0$ 时, $f_1(\tau)$ 和 $f_2(t-\tau)$ 的图如下:



$$\begin{aligned} y(t) &= f_1(t) * f_2(t) \\ &= \int_{-\infty}^{+\infty} f_1(\tau) f_2(t-\tau) d\tau \\ &= \int_{-\infty}^1 1 \cdot e^{-(t-\tau+1)} d\tau + \int_1^{t+1} 2 \cdot e^{-(t-\tau+1)} d\tau \end{aligned}$$

$$\begin{aligned}
 &= \int_{-\infty}^{-t} e^u du + \int_{-t}^0 2 \cdot e^u du \\
 &= e^{-t} + 2 - 2e^{-t} \\
 &= 2 - e^{-t}
 \end{aligned}$$

$$y(t) = \begin{cases} 1 & , t \leq 0 \\ 2 - e^{-t} & , t > 0 \end{cases}$$



5-2

$$(1) \quad R_X(0) \geq R_X(\tau)$$

$$R_X(\tau) = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} x(t) x(t-\tau) dt$$

$$\begin{aligned}
 R_{X^2}(\tau) &= E^2[x(t)x(t-\tau)] \leq E[x^2(t)]E[x^2(t-\tau)] = E[x^2(t)]E[x^2(t-0)] \\
 &= R_X(0)R_X(0) \\
 &= R_X^2(0)
 \end{aligned}$$

$$\therefore R_{X^2}(\tau) \leq R_{X^2}(0)$$

$$(2) \quad \rho_X(\tau) = \frac{E[(x(t) - \mu_X)(x(t-\tau) - \mu_X)]}{\sigma_X^2} = \frac{R_X(\tau) - \mu_X^2}{\sigma_X^2}$$

$$|\rho_X| \leq 1$$

$$\left| \frac{R_X(\tau) - \mu_X^2}{\sigma_X^2} \right| \leq 1$$

$$\Rightarrow \sigma_X^2 - \mu_X^2 \leq R_X(\tau) \leq \sigma_X^2 + \mu_X^2$$

$$(3) R_x(\tau) = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} x(t) x(t-\tau) dt = E[x(t) \cdot x(t-\tau)]$$

$$R_x(-\tau) = E[x(t) \cdot x(t+\tau)]$$

$$\text{let } t' = t + \tau$$

$$R_x(-\tau) = E[x(t'-\tau) \cdot x(t')] = E[x(t) \cdot x(t-\tau)] = R_x(\tau)$$

5-3

$$(1) \rho_{xy}(\tau) = \frac{E[(x(t) - \mu_x)(y(t-\tau) - \mu_y)]}{\sigma_x \sigma_y} = \frac{R_{xy}(\tau) - \mu_x \mu_y}{\sigma_x \sigma_y}$$

$$|\rho_{xy}(\tau)| \leq 1$$

$$\left| \frac{R_{xy}(\tau) - \mu_x \mu_y}{\sigma_x \sigma_y} \right| \leq 1$$

$$\Rightarrow \mu_x \mu_y - \sigma_x \sigma_y \leq R_{xy}(\tau) \leq \mu_x \mu_y + \sigma_x \sigma_y$$

$$(2) R_{yx}(-\tau) = E[x(t) y(t+\tau)]$$

$$\text{let } t' = t + \tau$$

$$= E[x(t'-\tau) y(t')]$$

$$= E[y(t) x(t-\tau)] = R_{yx}(\tau)$$